PARTICLE MODEL OF MATTER: TEACHER GUIDE

Eight 50-minute lessons over four weeks, for AQA Physics 4.3 (with 4.1.1.3) and Combined Science 6.3 (with 6.1.1.3). Each lesson has its own booklet in the student workbook and its own section of the teaching deck.

HOW THE RESOURCES FIT TOGETHER

RESOURCE	USED FOR
Teaching deck	52 slides with revision notes (press N). Slides 3–46 are for every class. Slides 47–52, after the Key words slide, are flagged **Triple Science only**: pressure and volume ($pV = \text{constant}$) and doing work on a gas.
Student workbook	One booklet per lesson (12 pt text, wide writing lines), each with the toolkit so far, its key words and quick answers. Every lesson: Do now, explanations, worked examples with "Your turn", graded practice, practical records, exam questions and a revision box.
Answer edition	The same pages and page numbers as each booklet, with answers in blue. For marking, or for students to mark their own work.
Unit review	39 marks of mixed exam practice, a fix-it log, the toolkit, all the key words and a progress tracker. Its last pages are the Triple Science only extension, matching deck slides 47–52.
Preps	Four one-page homework sheets, each out of 10, set after Lessons 2, 4, 6 and 8: three Do Now bank questions on earlier learning, then questions on the two latest lessons.

DESIGN PRINCIPLES

- **One model, used again and again.** Lesson 1 sets up the particle diagrams; every later lesson explains its new idea (pressure, density, internal energy, latent heat) with the same particles.
- **Worked example, your turn, faded practice.** Calculations go from a full frame (Equation, Substitute, Answer) to no frame to a challenge. The equation is always written in symbols before any number is substituted.
- **Every blank shows its options.** Gap-fill sentences print the choices after the gap, and ring boxes follow the AQA format.
- **Both required practicals have their own lesson**, with the workbook as the lab sheet. The specific heat capacity practical measures energy with an ammeter, voltmeter and stopwatch ($E = VIt$), which revisits Electric Circuits; the joulemeter is shown as the alternative, because exam questions use either.
- **Retrieval every lesson.** "Do now" questions revisit earlier lessons and Electric Circuits; "Revise it" gives cover-and-answer questions with quick answers at the back.
- **Equations are still taught for understanding.** AQA provides the full Physics Equations Sheet in every paper, which includes $\rho = m \div V$, $\Delta E = mc\Delta\theta$, $E = mL$ and $pV = \text{constant}$. Students still need the meaning of each symbol, the units and the rearrangements.
- **Higher-tier content is not marked.** Both courses are taught at Higher. Only Physics-only content is flagged, as "Triple only".

SEQUENCE

#	LESSON	WEEK	SLIDES
1	**Solids, liquids and gases** Particle diagrams, arrangement and motion, changes of state, mass is conserved	1	3–7
2	**Gas particles and pressure** Random motion, temperature and kinetic energy, what causes gas pressure, heating a gas at constant volume	1	8–12
3	**Density** $\rho = \frac{m}{V}$, units, density and the particle model, finding a mass or a volume	2	13–17
4	**Measuring density (required practical)** Regular solids, irregular solids by displacement, liquids; resolution and sources of error	2	18–23
5	**Internal energy and heating curves** Internal energy, the two things heating can do, heating and cooling graphs	3	24–28
6	**Specific latent heat** Latent heat, $E = mL$, fusion and vaporisation, measuring L for melting ice	3	29–34
7	**Specific heat capacity** What a temperature rise depends on, $\Delta E = mc\Delta\theta$, rearranging, heat capacity or latent heat?	4	35–39
8	**Measuring specific heat capacity (required practical)** Heating a metal block, $E = VIt$ or a joulemeter, the graph method, why the answer comes out too high	4	40–44

The Scheme of Work places P3.1–P3.3 in weeks 1–2 and P3.4–P3.5 in weeks 3–4. With both required practicals given a full lesson, internal energy (P3.3) is the first lesson of week 3. Triple Science classes need one extra lesson for slides 47–52 and the extension pages of the Unit review.

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Lesson 01

SOLIDS, LIQUIDS AND GASES

AQA Physics 4.3.1.1, 4.3.1.2; Combined Science 6.3.1.1, 6.3.1.2; 50 minutes; deck slides 3–7

BIG QUESTION **An ice cube, a puddle and the steam from a kettle are all water. What is different about them?**

STUDENTS WILL BE ABLE TO

- draw particle diagrams for a solid, a liquid and a gas
- describe how the particles are arranged and how they move
- name the six changes of state
- explain why mass is conserved when a substance changes state

WHY THIS LESSON, HERE

Every later lesson explains something with particle diagrams, so the diagrams and the vocabulary come first. Students met the three states at Key Stage 3; the new demands are precise wording ("vibrate about fixed positions"), correct diagrams, and using conservation of particles to explain conservation of mass.

PREPARATION AND EQUIPMENT

Two 20 cm³ syringes with the nozzles sealed (one full of air, one full of water). A beaker of crushed ice covered with cling film on a top-pan balance, set up at the start so it melts during the lesson (stand it in warm water for a few minutes if it is slow, then dry the outside). Deck slides 3–7.

LESSON SEQUENCE

MIN	PHASE	TEACHER MOVES (WORKBOOK QUESTION NUMBERS IN THE LESSON)
0–6	**Starting point**	The three 'What do you already know?' boxes. Read the ice mass on the balance and write it on the board. Pose the big question.
6–18	**Learn: three states**	Slides 4–5. Students compare the three diagrams and the table. Model drawing a liquid on the board: touching, no pattern. Syringe demonstration (Q7).
18–28	**Learn: changes of state**	Slide 6. Name the six changes; separate boiling from evaporating. Slide 7: mass is conserved, physical change.
28–42	**Try it**	Q1–Q5. Circulate on Q1 and Q4. Q6 is the challenge.
42–46	**Melted ice**	Read the balance again; students complete Q8–Q9.
46–50	**Exam corner**	Q10 as the hinge. Q11 for homework if short of time.

! **Hinge check.** Mini whiteboards: "Draw nine particles of a liquid." Any drawing with gaps between the particles, or with rows, needs the slide 4 model again before Lesson 3 (density depends on it).

MISCONCEPTIONS TO EXPECT

STUDENTS MAY THINK…	ADDRESS IT BY…
The particles in a liquid are spread out, halfway between a solid and a gas.	Show that a liquid cannot be squashed (syringe): its particles are touching. Draw the liquid by 'shaking' the solid diagram, not by spacing it out.
Particles melt, expand or get hot themselves.	The particles do not change. Only their arrangement and motion change. Insist on "the substance melts", never "the particles melt".
Mass is lost when ice melts, or when water boils.	The balance demonstration. For boiling, ask where the particles are now: in the room.
Evaporating needs the liquid to be at its boiling point.	Puddles dry up at 10 °C. Evaporating is from the surface at any temperature.

SUPPORT

Give the comparison table as a card sort (three states × four rows). Offer sentence starters for Q10: "In a solid the particles are… In a gas the particles are…".

STRETCH

Q6. Then: iodine sublimates when warmed gently; predict what you would see and what happens on a cold surface held above it.

Homework: Workbook: Revise it, then check with the quick answers at the back.

Lesson

02 GAS PARTICLES AND PRESSURE

AQA Physics 4.3.3.1; Combined Science 6.3.3.1; 50 minutes; deck slides 8–12

BIG QUESTION **An aerosol can says “Do not expose to temperatures above 50 °C”. What could go wrong?**

STUDENTS WILL BE ABLE TO

- describe the motion of the molecules in a gas
- link the temperature of a gas to the average kinetic energy of its molecules
- explain how a gas pushes on the walls of its container
- explain why heating a gas at constant volume increases its pressure

WHY THIS LESSON, HERE

The Scheme of Work pairs the particle model with motion in gases (P3.1). Gas pressure is the first place students must build a causal chain from particle behaviour to something they can measure, the skill that 3-mark ‘explain’ questions test throughout the unit. Only the qualitative link at constant volume is needed for Combined Science.

PREPARATION AND EQUIPMENT

A round-bottomed flask sealed with a bung and tube to a Bourdon gauge or pressure sensor; a large beaker or water bath; kettle; ice; thermometer. Test for leaks first. If you have no gauge, use the sample results (20 °C, 101 kPa; 40 °C, 108 kPa; 60 °C, 115 kPa; 80 °C, 122 kPa). Deck slides 8–12; slide 9 has a live model with temperature steps.

LESSON SEQUENCE

MIN	PHASE	TEACHER MOVES (WORKBOOK QUESTION NUMBERS IN THE LESSON)
0–6	**Do now**	Retrieval from Lesson 1 and Year 9 kinetic energy. Then the aerosol can warning: take predictions.
6–16	**Learn: random motion**	Slide 9: run the model, step the temperature up and down. Draw out ‘random’ and ‘average’ kinetic energy.
16–26	**Learn: pressure**	Slides 10–11. Build the four-step chain on the board with the class, then cover it and have pairs rebuild it aloud.
26–36	**Demonstration**	Flask and gauge in iced, room-temperature and hot water. Students record in Q6–Q8.
36–46	**Try it**	Q1–Q4; Q5 is the challenge. Q3 (ordering) checks the chain.
46–50	**Exam corner**	Q9 under timed conditions (3 minutes), then mark against the three steps.

! **Hinge check.** “A sealed can is cooled. Which is true? A: the particles get smaller. B: the particles hit the walls less often and less hard. C: some particles leave.” Anything but B: return to slide 10.

MISCONCEPTIONS TO EXPECT

STUDENTS MAY THINK…	ADDRESS IT BY…
Heating makes the particles expand.	The particles stay the same size and move faster. Ban “the particles expand”.
Pressure is caused by particles hitting each other.	Only collisions with the walls push on the walls. Ask what the gauge is connected to.
Hot gas particles all move at the same speed.	The model on slide 9 shows a spread of speeds: hence ‘average’ kinetic energy.

SUPPORT

Give the four steps of the chain on cards to sequence before writing Q4. Accept bullet points for the explanation.

STRETCH

Q5. Triple Science: preview slide 48 by asking what would happen to the pressure if the same gas were squeezed into half the volume.

Homework: Prep 1 (Lessons 1–2).

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Lesson

03 DENSITY

AQA Physics 4.3.1.1; Combined Science 6.3.1.1; 50 minutes; deck slides 13–17

BIG QUESTION **A gold bar the size of a carton of milk is too heavy for most people to lift with one hand. Why?**

STUDENTS WILL BE ABLE TO

- calculate density with $\rho = \frac{m}{V}$
- use the units kg/m³ and g/cm³
- explain differences in density using the particle model
- rearrange the equation to find a mass or a volume

WHY THIS LESSON, HERE

Density is the unit's first calculation and its first required practical. The particle model gives it meaning (how much mass is packed into each cubic centimetre) before the equation appears. Both unit systems are taught now, because the practical next lesson works in g and cm³ while exam questions often give kg and m³.

PREPARATION AND EQUIPMENT

A set of same-size blocks of different materials to pass round (density cubes are ideal). Calculators. Deck slides 13–17.

LESSON SEQUENCE

MIN	PHASE	TEACHER MOVES (WORKBOOK QUESTION NUMBERS IN THE LESSON)
0–6	**Do now**	Includes a g to kg conversion, needed today.
6–12	**Hook**	Pass the blocks round: same size, different mass. Big question about the gold bar (a 1-litre gold bar has a mass of 19.3 kg).
12–26	**Learn: the equation**	Slide 14. Worked example, then 'Your turn' on mini whiteboards. Slide 16: the two sets of units; water as the anchor value.
26–32	**Learn: particles**	Slide 15. Link back to the Lesson 1 diagrams: count the particles in each box.
32–46	**Try it**	Q1–Q6. Q4 (table) needs rearranging: model the second worked example first (slide 17). Q7 is the challenge.
46–50	**Exam corner**	Q8 and Q10; Q9 for homework.

! **Hinge check.** "A block has a mass of 200 g and a volume of 50 cm³. Its density is: A 0.25 g/cm³, B 4 g/cm³, C 10 000 g/cm³." A means the fraction is upside down: reteach 'mass on top'.

MISCONCEPTIONS TO EXPECT

STUDENTS MAY THINK…	ADDRESS IT BY…
Dense means heavy.	A whole room of air has more mass than a gold ring. Density compares equal volumes.
Cutting a block in half halves its density.	Mass and volume both halve, so the ratio is unchanged. Do it with numbers.
Converting g/cm³ to kg/m³ means dividing by 1000 (because g to kg does).	1 m³ is a million cm³. Use water as the check: 1 g/cm³ must become 1000 kg/m³.
Gas particles are lighter than solid particles.	The same particles; they are further apart.

SUPPORT

Keep to one unit system at a time. The formula triangle beside the equation is there for Q4; require the rearranged equation to be written before substituting.

STRETCH

Q7. Then: estimate the mass of the air in the classroom from its dimensions (density of air 1.2 kg/m³).

Homework: Workbook: Revise it, then check with the quick answers.

Lesson

04 MEASURING DENSITY (REQUIRED PRACTICAL)

AQA Physics 4.3.1.1, required practical 5; Combined Science 6.3.1.1, required practical 17; 50 minutes; deck slides 18–23

BIG QUESTION **A jeweller is offered a lump of "gold" with an awkward shape. How could its density be measured?**

STUDENTS WILL BE ABLE TO

- measure the density of a regular solid from its dimensions
- measure the volume of an irregular solid by displacement
- measure the density of a liquid
- choose instruments and explain how to make the measurements accurate

WHY THIS LESSON, HERE

Required practical 5 (Physics) / 17 (Combined Science). Students need all three methods, and exam questions concentrate on displacement, on the choice of instrument and on sources of error. The lab comes before the questions in this booklet, because the questions use the students' own results.

PREPARATION AND EQUIPMENT

Per group: balance (0.1 g or better), ruler, vernier callipers, two or three regular blocks, two irregular objects on thread (stones, lumps of modelling clay), 100 cm³ measuring cylinder, displacement can and beaker, water, a second liquid (salt water: about 15 g of salt per 100 cm³), paper towels. Deck slides 18–23. A circus of three stations (A, B, C) works well with 10 minutes at each.

LESSON SEQUENCE

MIN	PHASE	TEACHER MOVES (WORKBOOK QUESTION NUMBERS IN THE LESSON)
0–5	**Do now**	Density equation, 1 ml = 1 cm³, volume of a cuboid.
5–13	**Brief**	Slides 19–21: one slide for each method. Demonstrate reading a meniscus at eye level and using the callipers. Safety: spills.
13–43	**Practical**	Parts A, B and C, about 10 minutes each (Q1–Q3). Insist that masses are measured before objects get wet.
43–50	**Analysis**	Q4 with their own results; slide 22 on errors. Q5–Q7 and the Exam corner are finished next lesson's Do now time or at home.

! **Hinge check.** Hold up a measuring cylinder with a stone in it. "The reading went from 50 to 68 cm³. What is the volume of the stone?" Then: "What if a bubble was stuck to it?" (measured volume too big, density too small).

MISCONCEPTIONS TO EXPECT

STUDENTS MAY THINK…	ADDRESS IT BY…
The volume of the stone is the final reading.	It is the change in the reading. Label the table columns aloud as you demonstrate.
Read the top edge of the water, where it touches the glass.	Sketch a meniscus on the board; the bottom of the curve is the level.
A more expensive instrument is always 'more accurate'.	Use 'resolution' for the smallest change an instrument shows. Callipers have a better resolution than a ruler; whether the result is accurate depends on how they are used.

SUPPORT

Give groups one object per part, and pre-drawn sample data for any part they cannot finish. Pair less confident students for Part C, which has the most subtraction.

STRETCH

Q7 (the floating cork). Then compare the density of a regular block found from its dimensions with the value found by displacement, and explain which they trust more.

Homework: Prep 2 (Lessons 3–4), and the 6-mark method question (Q8) if not done in class.

Lesson

05 INTERNAL ENERGY AND HEATING CURVES

AQA Physics 4.3.2.1, 4.3.2.3; Combined Science 6.3.2.1, 6.3.2.3; 50 minutes; deck slides 24–28

BIG QUESTION **A pan of water boils on the hob for ten minutes. The hob keeps supplying energy, but the water never gets hotter than 100 °C. Where does the energy go?**

STUDENTS WILL BE ABLE TO

- state what internal energy is
- describe the two things that can happen when a substance is heated
- explain the flat sections of a heating or cooling graph
- read melting and boiling points from a graph

WHY THIS LESSON, HERE

This is the conceptual centre of the unit: heating does one of two things, and the particle model says which energy changes. Flat sections on heating graphs are counter-intuitive ("the heater is on, why is it not getting hotter?"), so students collect or plot a cooling curve themselves before latent heat is quantified in Lesson 6.

PREPARATION AND EQUIPMENT

Per group: a boiling tube one third full of stearic acid, melted in a water bath at about 90 °C before the lesson, a thermometer in each tube, clamp stands, stopwatches, eye protection. The class practical takes 12 minutes of readings; if time is short, run one tube as a demonstration under a visualiser and give the sample results. Deck slides 24–28.

LESSON SEQUENCE

MIN	PHASE	TEACHER MOVES (WORKBOOK QUESTION NUMBERS IN THE LESSON)
0–5	**Do now**	Then start the cooling immediately: tubes out of the bath, first reading (Q5). One student in each group reads every minute while the lesson continues.
5–17	**Learn: internal energy**	Slides 25–26. Kinetic and potential energy of particles; the two outcomes of heating. Keep recording.
17–27	**Learn: graphs**	Slides 27–28. Label the heating graph for water with the class.
27–39	**Plot**	Q6: plot the cooling curve from their readings; Q7–Q8.
39–46	**Try it**	Q1–Q3; Q4 is the challenge.
46–50	**Exam corner**	Q9, marked against the three points.

! **Hinge check.** Show the heating graph. "During the flat section, the energy of the particles is: A not changing, B increasing as kinetic energy, C increasing as potential energy." A is the common answer: heating has not stopped.

MISCONCEPTIONS TO EXPECT

STUDENTS MAY THINK…	ADDRESS IT BY…
On a flat section no energy is being supplied.	The heater is still on. Ask where the energy has gone: into separating the particles.
Temperature and internal energy are the same thing.	1 kg of steam and 1 kg of water at 100 °C: same temperature, different internal energy (Q4).
The potential energy here is gravitational.	It is the energy due to the particles' positions relative to each other. Pulling them apart needs energy, like stretching a spring.

SUPPORT

Give the graph axes with the scales already numbered. For Q2, have students colour the sloping sections one colour and the flat sections another before answering.

STRETCH

Q4. Then sketch the cooling graph for steam at 120 °C cooled to ice at −10 °C, with both flat sections labelled.

Homework: Workbook: Revise it, then check with the quick answers.

Lesson

06 SPECIFIC LATENT HEAT

AQA Physics 4.3.2.3; Combined Science 6.3.2.3; 50 minutes; deck slides 29–34

BIG QUESTION **A scald from steam at 100 °C is far worse than a scald from water at 100 °C. Why?**

STUDENTS WILL BE ABLE TO

- state what specific latent heat means
- calculate with $E = mL$
- tell fusion from vaporisation
- describe an experiment to measure the specific latent heat of fusion of water

WHY THIS LESSON, HERE

Lesson 5 established that a change of state needs energy at constant temperature; now it is measured. $E = mL$ has no temperature term, which makes it the simpler of the two heating equations, so it comes before specific heat capacity. The specification also asks for an experiment to measure the latent heat of fusion of water.

PREPARATION AND EQUIPMENT

Demonstration or groups: two filter funnels of crushed ice (taken from the freezer 15 minutes earlier so it is at 0 °C and wet), two beakers, 12 V immersion heater, joulemeter, power supply, balance. Run the heater for 5 minutes; start it early in the lesson. Deck slides 29–34.

LESSON SEQUENCE

MIN	PHASE	TEACHER MOVES (WORKBOOK QUESTION NUMBERS IN THE LESSON)
0–6	**Do now**	Then set the ice experiment running (slide 33) so that it melts during the teaching.
6–18	**Learn**	Slides 30–31: latent heat, the definition, $E = mL$. Worked example, ‘Your turn’ on mini whiteboards.
18–24	**Learn: two kinds**	Slide 32: fusion and vaporisation, the values for water, and the steam scald.
24–36	**Try it**	Q1–Q4; Q5 is the challenge. Model the second pair of worked examples (rearranging) before Q4.
36–46	**Practical results**	Read the joulemeter and both masses; Q6–Q8. Compare the class value with 334 000 J/kg.
46–50	**Exam corner**	Q9 (grams to kilograms) as the hinge.

! **Hinge check.** “Which of these needs $E = mL$? A: warming water from 20 °C to 30 °C. B: melting ice at 0 °C. C: cooling a block of iron.” Then ask why there is no temperature in the equation.

MISCONCEPTIONS TO EXPECT

STUDENTS MAY THINK…	ADDRESS IT BY…
Boiling water keeps getting hotter the longer it boils.	A thermometer in a boiling kettle or beaker stays at 100 °C.
Latent heat is only taken in, never given out.	The same energy is released on freezing and condensing: the steam scald, and why frost-protection sprays work.
‘Fusion’ means joining nuclei.	In this topic it is the old word for melting.

SUPPORT

Keep masses in kilograms for the first questions. Provide the frame (Equation, Substitute, Answer) for every calculation.

STRETCH

Q5. Then: why does the control funnel need the same mass of ice and the same room? What would the result be without it (too much melted, so L too small)?

Homework: Prep 3 (Lessons 5–6).

Lesson 07

SPECIFIC HEAT CAPACITY

AQA Physics 4.3.2.2, 4.1.1.3; Combined Science 6.3.2.2, 6.1.1.3; 50 minutes; deck slides 35–39

BIG QUESTION **On a sunny beach the sand burns your feet but the sea is still cold. Both have had the same sunshine. Why?**

STUDENTS WILL BE ABLE TO

- state the three things a temperature rise depends on
- state what specific heat capacity means
- calculate with $\Delta E = m c \Delta\theta$
- decide whether a problem needs specific heat capacity or specific latent heat

WHY THIS LESSON, HERE

Specific heat capacity appears in both the Energy section (4.1.1.3) and here (4.3.2.2); the Year 9 Energy unit left it for this unit. It has the unit's hardest equation, with three variables to rearrange and a temperature *change* to calculate first. Ending with 'heat capacity or latent heat?' makes students choose between the two equations, which is how exam questions test them.

PREPARATION AND EQUIPMENT

Two identical hotplates or heaters; 200 g of water and 200 g of cooking oil in matching beakers, each with a thermometer (demonstration: equal energy, different temperature rises). Calculators. Deck slides 35–39.

LESSON SEQUENCE

MIN	PHASE	TEACHER MOVES (WORKBOOK QUESTION NUMBERS IN THE LESSON)
0–6	**Do now**	Includes a temperature change, the step most often missed.
6–12	**Demonstration**	Heat the water and the oil for the same time. Predict first; read both thermometers. Slide 36.
12–26	**Learn**	Slide 37: definition and $\Delta E = m c \Delta\theta$. Worked example; 'Your turn'. Slide 38: rearranging.
26–42	**Try it**	Q1–Q4. Q5 (which equation?) after slide 39. Q6 is the two-step challenge.
42–50	**Exam corner**	Q7 and Q8 timed; Q9 for homework.

! **Hinge check.** "Water is heated from 20 °C to 100 °C. $\Delta\theta$ is: A 100 °C, B 80 °C, C 120 °C." Then: "Which equation for melting ice at 0 °C?"

MISCONCEPTIONS TO EXPECT

STUDENTS MAY THINK…	ADDRESS IT BY…
$\Delta\theta$ is the final temperature.	It is the change. Make students write the subtraction as a line of working every time.
A high specific heat capacity means it heats up quickly.	The opposite: more energy is needed for each degree. Water against oil in the demonstration.
Heat and temperature are the same.	The same energy gave two different temperature rises.
The mass can stay in grams.	c is per kilogram. Convert first (Q3).

SUPPORT

Use the five-row frame (Convert, Temperature change, Equation, Substitute, Answer). The formula triangle helps with Q4.

STRETCH

Q6. Then: 0.10 kg of ice at 0 °C is melted and warmed to 20 °C; calculate the total energy (33 400 J + 8400 J).

Homework: Workbook: Revise it, then check with the quick answers.

Lesson

08 MEASURING SPECIFIC HEAT CAPACITY (REQUIRED PRACTICAL)

AQA Physics 4.1.1.3, required practical 1; Combined Science 6.1.1.3, required practical 14; 50 minutes; deck slides 40–44

BIG QUESTION How could you find out which metal a block is made of, using only a heater and a thermometer?

STUDENTS WILL BE ABLE TO

- describe how to measure the specific heat capacity of a metal block
- calculate the energy supplied with $E = VIt$, or read it from a joulemeter
- calculate the specific heat capacity from the results
- explain why the result is usually too high, and how to improve it

WHY THIS LESSON, HERE

Required practical 1 (Physics) / 14 (Combined Science). It is the most examined practical in the Energy and Particle model sections, and the evaluation (why the result is too high) is as important as the method. Measuring the energy with an ammeter, a voltmeter and a stopwatch revisits $P = VI$ and $E = Pt$ from Electric Circuits; the joulemeter is the alternative, and students should recognise both in exam questions.

PREPARATION AND EQUIPMENT

Per group: 1 kg aluminium (or other) block with two holes, 12 V immersion heater, thermometer, insulation jacket, heatproof mat, power supply, ammeter (0–10 A), voltmeter, leads, stopwatch, a drop of oil for the thermometer hole. A balance to share. A joulemeter for at least one group if you have one, so the class sees both methods. Check each heater is rated for the supply. Deck slides 40–44.

LESSON SEQUENCE

MIN	PHASE	TEACHER MOVES (WORKBOOK QUESTION NUMBERS IN THE LESSON)
0–5	**Do now**	Includes $P = VI$ and $E = Pt$, which combine to give today's equation.
5–13	**Brief**	Slide 41: the method, both ways of measuring energy. Safety: the heater must be in the block before it is switched on.
13–31	**Practical**	Q1–Q2: set up, then 10 minutes of heating with readings every 2 minutes. During the wait, groups calculate the energy row.
31–43	**Results**	Q3 (from the totals), then Q4–Q6 (graph and gradient). If time is short, set the graph as homework and do Q3 and Q7 in class.
43–50	**Evaluate**	Slide 43; Q7. Compare class values with the accepted value.

! **Hinge check.** “Our value came out higher than the true one. Is that because the temperature rise we measured was too big or too small?” (Too small: energy went to the surroundings.)

MISCONCEPTIONS TO EXPECT

STUDENTS MAY THINK…	ADDRESS IT BY…
Energy losses make the answer too low.	Less energy reaches the block, so $\Delta\theta$ is too small, and dividing by a smaller number gives a bigger c.
Take the final temperature the moment the heater is switched off.	The temperature keeps rising for a minute as energy spreads through the block: record the highest reading.
The difference from the true value is ‘human error’.	Name the cause: energy transferred to the surroundings, a systematic error. Insulation reduces it.

SUPPORT

Give the energy row already calculated, so the group concentrates on the readings and on Q3. A partly drawn graph (axes and scales) helps with Q4.

STRETCH

Q10. Then: estimate the percentage of the energy supplied that did not stay in the block, using the accepted value.

Homework: Prep 4 (Lessons 7–8). Then the Unit review booklet before the end-of-unit test.